Prepare Community Data for Model Fitting

Description

Reshapes long-format community data into a wide matrix suitable for model fitting, filtering to the canonical sample index supplied by 'align_sample_ids()'.

Usage

```
prepare_community_for_fit(data_community_long = NULL, data_sample_ids = NULL)
```

Arguments

`data_community_long`	A data frame in long format with columns 'dataset_name', 'age', 'taxon', and 'value'.
`data_sample_ids`	A data frame of valid '(dataset_name, age)' pairs as returned by 'align_sample_ids()'.

Details

Only samples present in 'data_sample_ids' are retained. Rows with 'NA' or zero 'value' are dropped before pivoting. The function widens the data and converts the result to a matrix directly, so the output is ready for 'filter_constant_taxa()' and 'build_jsdm_fit_input()'.

Value

A numeric matrix with row names in the format '"<dataset_name>__<age>"' and taxon names as column names. Missing taxon–sample combinations are filled with '0'. Rows are ordered by 'dataset_name' then 'age', matching the order in 'data_sample_ids'.

See Also

[align_sample_ids()], [filter_constant_taxa()], [build_jsdm_fit_input()]